

TASK RECORD

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COLLEGE: Vels institute of science, Technology and Advanced studies

INTERNSHIP ROLE: Data Analyst

1. TASK: Linux user management commands, basic system administration:

Task Description:

The objective of this task is to learn and practice basic Linux user management and file system commands. The task includes creating a new user, setting a password, modifying user information, navigating directories, viewing user details, displaying command history, and understanding the Linux file system. These commands help users gain practical knowledge of Linux system administration and command-line operations.

Commands / Tools Used:

Pwd, whoami, ls, cd, cat /etc/passwd, sudo useradd -m <username>, sudo passwd <username>, sudo usermod -d <directory> -m <username>

Commands practiced:

- pwd – Display the current working directory.
- whoami – Display the current logged-in user.
- ls – List files and directories.
- cd – Change the current directory.
- history – Display previously executed commands.
- cat /etc/passwd – View user account information.
- sudo useradd -m <username> – Create a new user with a home directory.
- sudo passwd <username> – Set or change the user's password.

- `sudo usermod -d <directory> -m <username>` – Modify the user's home directory.

OUTPUT:

```
vikasini@LAPTOP-DPLUR7N6:~$ whoami
vikasini
vikasini@LAPTOP-DPLUR7N6:~$ pwd
/home/vikasini
vikasini@LAPTOP-DPLUR7N6:~$ ls
chmod_practice  new-folder  ubuntupractice
vikasini@LAPTOP-DPLUR7N6:~$ cd ..
vikasini@LAPTOP-DPLUR7N6:~$ pwd
/home
vikasini@LAPTOP-DPLUR7N6:~$ cd ~
vikasini@LAPTOP-DPLUR7N6:~$ pwd
/home/vikasini
vikasini@LAPTOP-DPLUR7N6:~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
systemd-timesync:x:996:996:systemd Time Synchronization:/:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/false
messagebus:x:101:101::/nonexistent:/usr/sbin/nologin
syslog:x:102:102::/nonexistent:/usr/sbin/nologin
systemd-resolve:x:991:991:systemd Resolver:/:/usr/sbin/nologin
```

```
WHOAMI(1)                                User Commands                                WHOAMI(1)

NAME
    whoami - print effective user name

SYNOPSIS
    whoami [OPTION]...

DESCRIPTION
    Print the user name associated with the current effective user ID.  Same as id -un.

    --help display this help and exit

    --version
        output version information and exit

AUTHOR
    Written by Richard Mlynarik.

REPORTING BUGS
    GNU coreutils online help: <https://www.gnu.org/software/coreutils/>
    Report any translation bugs to <https://translationproject.org/team/>

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    This is free software: you are free to change and redistribute it.  There is NO WARRANTY, to the extent permitted by law.

SEE ALSO
    Full documentation <https://www.gnu.org/software/coreutils/whoami>
    or available locally via: info '(coreutils) whoami invocation'

GNU coreutils 9.4                        April 2024                        WHOAMI(1)
Manual page whoami(1) line 1/33 (END) (press h for help or q to quit)
```

```
root@LAPTOP-DPLUR7N6: /h  X + v
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
systemd-timesync:x:996:996:systemd Time Synchronization:/:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/false
messagebus:x:101:101::/nonexistent:/usr/sbin/nologin
syslog:x:102:102::/nonexistent:/usr/sbin/nologin
systemd-resolve:x:991:991:systemd Resolver:/:/usr/sbin/nologin
uidd:x:103:103::/run/uidd:/usr/sbin/nologin
landscape:x:104:105::/var/lib/landscape:/usr/sbin/nologin
polkitd:x:990:990:User for polkitd:/:/usr/sbin/nologin
vikasini:x:1000:1000::,/home/vikasini:/bin/bash
vikasini@LAPTOP-DPLUR7N6:~$ man whoami
vikasini@LAPTOP-DPLUR7N6:~$ man pwd
vikasini@LAPTOP-DPLUR7N6:~$ whoami
vikasini
vikasini@LAPTOP-DPLUR7N6:~$ sudo su
```

```
vikasini@LAPTOP-DPLUR7N6: X + v
vikasini@LAPTOP-DPLUR7N6:~$ pwd
/home/vikasini
vikasini@LAPTOP-DPLUR7N6:~$ sudo useradd -m student
vikasini@LAPTOP-DPLUR7N6:~$ sudo passwd student
New password:
Retype new password:
passwd: password updated successfully
vikasini@LAPTOP-DPLUR7N6:~$ sudo usermod -d /home/ -m student
usermod: directory /home exists
vikasini@LAPTOP-DPLUR7N6:~$ ls /home
student  vikasini
vikasini@LAPTOP-DPLUR7N6:~$ cd /home/student
-bash: cd: /home/student: Permission denied
vikasini@LAPTOP-DPLUR7N6:~$ pwd
/home/vikasini
vikasini@LAPTOP-DPLUR7N6:~$ |
```

```
vikasini@LAPTOP-DPLUR7N6: X + v
vikasini@LAPTOP-DPLUR7N6:~$ pwd
/home/vikasini
vikasini@LAPTOP-DPLUR7N6:~$ sudo useradd -m student
vikasini@LAPTOP-DPLUR7N6:~$ sudo passwd student
New password:
Retype new password:
passwd: password updated successfully
vikasini@LAPTOP-DPLUR7N6:~$ sudo usermod -d /home/ -m student
usermod: directory /home exists
vikasini@LAPTOP-DPLUR7N6:~$ ls /home
student vikasini
vikasini@LAPTOP-DPLUR7N6:~$ cd /home/student
-bash: cd: /home/student: Permission denied
vikasini@LAPTOP-DPLUR7N6:~$ pwd
/home/vikasini
vikasini@LAPTOP-DPLUR7N6:~$ |
```

```
vikasini@LAPTOP-DPLUR7N6: X + v
199 sudo passwd student1
200 sudo usermod -d /home/student1 -m student1
201 sudo userdel -r student1
202 clear
203 whoami
204 pwd
205 ls
206 cd ..
207 pwd
208 cd ~
209 pwd
210 cat /etc/passwd
211 man whoami
212 man pwd
213 whoami
214 sudo su
215 sudo usermod -d /home/ -m student1
216 cat /etc/passwd
217 ls /home
218 clear
219 sudo usermod -d /home/ -m vikasini
220 sudo usermod -d /home/ -m student1
221 sudo usermod -d /home/ -m student
222 whoami
223 man whoami
224 clear
225 pwd
226 sudo useradd -m student
227 sudo passwd student
228 sudo usermod -d /home/ -m student
229 ls /home
230 cd /home/student
231 pwd
232 history
233 clear
234 history
vikasini@LAPTOP-DPLUR7N6:~$ |
```

PROBLEMS FACED:

1. Initially, there was confusion about the correct sequence of Linux commands.
2. Errors occurred because the specified user did not exist before running the usermod command.
3. Faced a "Permission denied" error while trying to access another user's home directory.
4. Received the message "user is currently used by process" when attempting to modify the home directory of the currently logged-in user.
5. Had difficulty understanding the purpose and usage of Linux user management commands.
6. Needed to switch between normal user mode and root mode to execute administrative commands.
7. Faced minor issues while setting passwords and understanding terminal outputs.

CONCLUSION:

Successfully learned and executed basic Linux commands and user management operations such as creating users, changing passwords, navigating directories, and viewing user information. This task improved understanding of the Linux operating system and command-line usage.